RV College of Engineering®
(An Autonomous Institution Affiliated to VTU)
Mysore Road, RV Vidyaniketan Post, Bengaluru – 560059, Karnataka, India
Go, change the world®

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT
OPERATIONS MANAGEMENT

Course Code: IM352IA • Semester: V • UG

Question Bank: Unit I and Unit II (CIE and SEE)

Scope and layout.

- **Unit I** – Using Operations to Compete. **Unit II** – Developing a Process Strategy and Planning Capacity (syllabus, as in the unit notes).
- **Part I (CIE):** every Unit I/II question from CIE-I (06/11/2025 and 25/11/2024), CIE-II (07/01/2025) and the Improvement CIE (12/01/2026). The CIE-II-A (15/12/2025) and CIE-III (27/01/2025) papers contain no Unit I/II questions and are left out. Questions repeated across sets are listed once, with every source given.
- **Part II (SEE):** following the SEE pattern, **Question 2** of each paper is Unit I and **Questions 3 and 4** are Unit II. Q3 and Q4 are alternatives (answer any one full question from each pair). Part A of the SEE papers is not included.
- **M** = marks; **BT** = Bloom's taxonomy level; **CO** = course outcome. Shown only where the printed paper gives them; "–" means not printed.

PART I – CONTINUOUS INTERNAL EVALUATION (CIE)

Unit I – Using Operations to Compete

Part A (fill in the blanks / short answer; 1 mark each unless stated)

No.	Question	M	BT	CO
1	When Amazon decides where to locate its fulfillment centers, it is making a _____ level decision in operations management. *Source: CIE-I, 06/11/2025*	1	2	2
2	Decision-making models in operations include deterministic, probabilistic, and _____ models. *Source: CIE-I, 06/11/2025*	1	1	3
3	The Fourth Industrial Revolution is driven by technologies such as IoT, AI, and _____. *Source: CIE-I, 06/11/2025*	1	1	1
4	When a factory balances machine utilization and customer delivery times, it is managing the _____ responsiveness trade-off. *Source: CIE-I, 06/11/2025*	1	2	2
5	Walmart's use of RFID and data analytics to track inventory is an example of using _____ as a driver of supply chain performance. *Source: CIE-I, 06/11/2025*	1	2	3
6	Zara's ability to rapidly design and distribute new fashions demonstrates the competitive priority of _____. *Source: CIE-I, 06/11/2025*	1	2	2
7	Operations ensure the efficient use of _____ to meet customer demands. *Source: CIE-I, 25/11/2024*	1	1	1
8	A supply chain includes the flow of materials, information, and _____ across organizations. *Source: CIE-I, 25/11/2024*	1	1	1

No.	Question	M	BT	CO
9	Effective supply chain management ensures timely delivery and reduces _____. *Source: CIE-I, 25/11/2024*	1	1	1
10	Competitive priorities like cost, quality, and _____ determine how a company competes in the market. *Source: CIE-I, 25/11/2024*	1	1	1
11	The Fourth Industrial Revolution integrates technologies like IoT, _____, and robotics into operations. *Source: CIE-I, 25/11/2024*	1	1	1
12	Processes can be categorized as _____ or supporting based on their direct impact on the final product. *Source: CIE-I, 25/11/2024*	1	1	1
13	Differentiate between order winner and order qualifier. *Source: CIE-I, 25/11/2024*	2	2	2
14	A hospital is considering a new procedure to be offered at Rs. 200 per patient. The fixed cost per year would be Rs. 1,00,000, with total variable costs of Rs. 100 per patient. What is the break-even quantity for this service? *Source: CIE-I, 25/11/2024*	2	2	2

Part B

No.	Question	M	BT	CO
1	Define operations management. Explain the key terms involved in the definition of operations management with a schematic model of a production system. *Source: CIE-I, 06/11/2025; CIE-I, 25/11/2024 (same question, worded slightly differently)*	10	2	1
2	With suitable example, distinguish between manufacturing and service processes. *Source: CIE-I, 06/11/2025; CIE-I, 25/11/2024 (same question)*	10	2	1
3	A retailer must decide whether to build a small or a large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6, respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = $223,000) or to expand (payoff = $270,000). If a small facility is built and demand is low, there is no reason to expand and the payoff is $200,000. If a large facility is built and demand proves to be low, the choice is to do nothing ($40,000) or to stimulate demand through local advertising. The response to advertising may be either modest or sizable, with their probabilities estimated to be 0.3 and 0.7, respectively. If it is modest, the payoff is estimated to be only $20,000; the payoff grows to $220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is $800,000. Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative – building a small facility or building a large facility – has the higher expected payoff? *Source: CIE-I, 06/11/2025*	16	4	3

No.	Question	M	BT	CO
4	A manager is deciding whether to build a small or a large facility. Much depends on the future demand that the facility must serve, and demand may be small or large. The manager knows with certainty the payoffs that will result under each alternative, shown in the payoff table below. What is the best alternative for each decision rule: Maximin, Maximax, Laplace, Minimax Regret?	14	4	3

Alternative	Low demand	High demand
Small facility	200	270
Large facility	160	800
Do nothing	0	0

Source: CIE-I, 06/11/2025 (14 M); CIE-I, 25/11/2024 (10 M)

5	White Valley Ski Resort is planning the ski lift operation for its new ski resort. Management is trying to determine whether one or two lifts will be necessary; each lift can accommodate 250 people per day. Skiing normally occurs in the 14-week period from December to April, during which the lift will operate 7 days per week. The first lift will operate at 90 percent capacity if economic conditions are bad, the probability of which is believed to be about 0.3. During normal times the first lift will be utilized at 100 percent capacity, and the excess crowd will provide 50 percent utilization of the second lift. The probability of normal times is 0.5. Finally, if times are really good, the probability of which is 0.2, the utilization of the second lift will increase to 90 percent. The equivalent annual cost of installing a new lift, recognizing the time value of money and the lift's economic life, is \$50,000. The annual cost of installing two lifts is only \$90,000 if both are purchased at the same time. If used at all, each lift costs \$200,000 to operate, no matter how low or high its utilization rate. Lift tickets cost \$20 per customer per day. Should the resort purchase one lift or two? *Source: CIE-I, 25/11/2024*	20	4	3

Unit II – Developing a Process Strategy and Planning Capacity

Part A (fill in the blanks / short answer; 1 mark each)

No.	Question	M	BT	CO
1	Strategies for change include process improvement, process redesign, and _____ process. *Source: CIE-I, 06/11/2025*	1	1	1
2	Strategic fit refers to the alignment between business strategy and _____ strategy. *Source: CIE-I, 06/11/2025*	1	2	2
3	In manufacturing, the four process structures are job process, batch process, line process, and _____ process. *Source: CIE-I, 06/11/2025*	1	1	1
4	In a make-to-order process, production begins only after receiving a _____ from the customer. *Source: CIE-I, 06/11/2025*	1	1	1
5	In services, the process structure is primarily influenced by the level of _____ and customization. *Source: CIE-II, 07/01/2025*	1	1	2
6	In manufacturing, a _____ process is ideal for high-volume, standardized production. *Source: CIE-II, 07/01/2025*	1	2	2

No.	Question	M	BT	CO
7	The first step in a systematic approach to long-term capacity decisions is _____ future demand. *Source: CIE-II, 07/01/2025*	1	2	2
8	A systematic approach to capacity planning involves forecasting, identifying alternatives, and evaluating _____. *Source: CIE-II, 07/01/2025*	1	1	2
9	_____ ensures that operational capabilities are in line with the company's strategic goals. *Source: CIE-II, 07/01/2025*	1	2	3
10	A _____ shop deals with customized, low-volume production. *Source: Improvement CIE, 12/01/2026*	1	1	2
11	In mass production, the layout used is _____ layout. *Source: Improvement CIE, 12/01/2026*	1	1	2
12	The maximum output a system can produce is called its _____. *Source: Improvement CIE, 12/01/2026*	1	1	2
13	The long-term decision of choosing a geographic site for operations is called _____. *Source: Improvement CIE, 12/01/2026*	1	2	3
14	Continuous processes are suitable for _____ production. *Source: Improvement CIE, 12/01/2026*	1	2	3

Part B

No.	Question	M	BT	CO
1	Explain the major process strategy decisions and their implications for operations. *Source: CIE-II, 07/01/2025*	10	2	2
2	Richard Garber is the head designer for Matthews and Novak Design Company. Garber has been called in to design the layout for a newly constructed office building. From statistical samplings over the past three months, Garber developed the following closeness matrix for daily trips between the department's offices. **a.** If other factors are equal, which two offices should be located closest together? **b.** A figure shows an alternative layout for the department: top row C, F, A; bottom row B, E, D. What is the total weighted-distance score for this plan based on rectilinear distance, assuming that offices A and B are three units of distance apart? *Source: CIE-II, 07/01/2025*	10	3	2
3	Explain the concepts of economies and diseconomies of scale with appropriate examples. *Source: Improvement CIE, 12/01/2026*	10	2	2
4	Explain process reengineering and process improvement, highlighting their differences with suitable examples. *Source: Improvement CIE, 12/01/2026*	10	4	3

Department	A	B	C	D	E	F
A	–	25	90		105	185
B		–			125	125
C			–		25	
D				–		105
E					–	
F						–

No.	Question	M	BT	CO
5	Describe a systematic approach to process analysis with the help of a block diagram. *Source: Improvement CIE, 12/01/2026*	10	2	2
6	A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.	10	3	3

Item	Client X	Client Y
Annual demand forecast (copies)	2,000	6,000
Standard processing time (hours/copy)	0.5	0.7
Average lot size (copies per report)	20	30
Standard setup time (hours)	0.25	0.40

Source: Improvement CIE, 12/01/2026

PART II – SEMESTER END EXAMINATION (SEE)

Nine SEE papers are covered: Feb/Mar 2026, Feb/Mar 2025 (IM352IA); Apr/May 2024 (21IM53); Mar/Apr 2023 (18IM54); Jan/Feb 2021 (18IM54) and Jan/Feb 2021 (16IM55); Fast Track Oct 2020 (16IM55/12IM62); Nov/Dec 2019 (16IM55); Fast Track Jul 2019 (16IM55/12IM62).

Unit I – Question 2 of each paper

Feb/Mar 2026 (IM352IA)

No.	Question	M	BT	CO
2(a)	The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Variable costs are estimated at \$7 per unit produced and sold. Fixed costs are about \$56,000 per year. (i) If the selling price is set at \$25, how many units must be produced and sold to break even? (ii) Forecasted sales for the first year are 10,000 units if the price is reduced to \$15. With this pricing strategy, what would be the product's total contribution to profits in the first year?	8	3	1
2(b)	With suitable example, distinguish between manufacturing processes and service processes.	8	2	1

Feb/Mar 2025 (IM352IA)

No.	Question	M	BT	CO
2(a)	Identify nine competitive priorities used in operations strategy and explain how a consistent pattern of decisions can develop organizational capabilities.	10	2	1
2(b)	Describe the process view of operations in typical organizations with the help of a block diagram.	6	2	1

Apr/May 2024 (21IM53)

No.	Question	M	BT	CO
2(a)	Differentiate between competitive priorities and competitive capabilities with suitable examples.	8	2	1
2(b)	IKEA shows how important operations management is for its own success and the success of any type of organization. In this view identify some of the activities that IKEA's operations managers are involved in through highlighting some eight important aspects.	8	3	2

Mar/Apr 2023 (18IM54)

No.	Question	M	BT	CO
2(a)	Explain the term operations management and identify the three major functional areas of business organizations and briefly describe how they are interrelated.	8	–	–
2(b)	Describe the new trends and challenges in operations management.	8	–	–

Jan/Feb 2021 (18IM54)

No.	Question	M	BT	CO
2(a)	Consider amazon.com, whose website enjoys millions of "hits" each day and puts customers in touch with millions of services and products. What are amazon.com's competitive priorities and what should its operations strategy focus on?	6	–	–
2(b)	A retailer must decide whether to build a small or large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6 respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = $223,000) or to expand (payoff = $270,000). If the small facility is built and demand is low, there is no reason to expand and the payoff is $200,000. If a large facility is built and demand proves to be low, the choice is to do nothing ($40,000) or to stimulate maximum demand through local advertising. The response to advertising may be either modest or sizable, with probabilities estimated to be 0.3 and 0.7 respectively. If it is modest, the payoff is estimated to be only $20,000; the payoff grows to $220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is $800,000. Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative – building a small facility or building a large facility – has the higher expected payoff?	10	–	–

Jan/Feb 2021 (16IM55)

No.	Question	M	BT	CO
2(a)	Developing a customer-driven operations strategy begins with corporate strategy, which coordinates the firm's overall goals with its core processes. Identify and discuss the factors connecting corporate strategy and key operations management decisions with block diagram.	8	–	–
2(b)	Calculate the productivity for the following operations: (i) Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week. (ii) A team of workers makes 400 units of a product, which is sold in the market for $10 each. The accounting department reports that for this job the actual costs are $400 for labour, $1,000 for materials, and $300 for overhead.	8	–	–

Fast Track Oct 2020 (16IM55/12IM62)

No.	Question	M	BT	CO
2(a)	Define operations management. Explain with the help of a block diagram the process view of operations management and write how processes work.	6	–	–

No.	Question	M	BT	CO
2(b)	A manager is trying to decide whether to build a small, medium or large facility. Demand can be low, average or high with estimated probabilities being 0.25, 0.40 and 0.35 respectively. A small facility is expected to earn an after tax net present value of just \$18,000 if demand is low. If demand is average, the small facility is expected to earn \$75,000; it can be increased to medium size to earn a net present value of \$60,000. If demand is high, the small facility is expected to earn \$75,000 and can be expanded to medium size to earn \$60,000 or to large size to earn \$125,000; a medium sized facility is expected to lose an estimated \$25,000 if demand is low and earn \$140,000 if demand is average. If the demand is high the medium sized facility is expected to earn a net present value of \$150,000; it can be expanded to a large size for a net pay off of \$145,000. If a large facility is built and demand is high, earnings are expected to be \$220,000. If demand is average for the large facility the present value is expected to be \$125,000; if demand is low, the facility is expected to lose \$60,000. (i) Draw a decision tree for this problem. (ii) What should the management do to achieve the highest expected pay off?	10	–	–

Nov/Dec 2019 (16IM55)

No.	Question	M	BT	CO
2(a)	Developing a customer-driven operations strategy begins with corporate strategy, which coordinates the firm's overall goals with its core processes. Identify and discuss the factors connecting corporate strategy and key operations management decisions with block diagram.	8	–	–
2(b)	Calculate the productivity for the following operations: (i) Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week. (ii) A team of workers makes 400 units of a product, which is sold in the market for \$10 each. The accounting department reports that for this job the actual costs are \$400 for labour, \$1000 for materials, and \$300 for overhead.	8	–	–

Fast Track Jul 2019 (16IM55/12IM62)

No.	Question	M	BT	CO
2(a)	How does the process view and the supply chain view of operations management help us understand the need and importance of operations management as a strategic function in the organization? Elaborate on these two perspectives of operations management.	8	–	–
2(b)	The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Before commercializing the device, the owner wants reasonable assurance of success. Variable costs are \$7 per unit produced and sold. Fixed costs are about \$500,000 per year. (i) If the price is reduced to \$15 and forecasted sales for the first year are 10,000 units, what would be the product's total contribution to profits in the first year? (ii) If the selling price is set at \$25, how many units must be produced and sold to break even? Use both algebraic and graphical approaches.	8	–	–

Unit II – Questions 3 and 4 of each paper

Q3 and Q4 are alternatives within each paper: answer any one full question from the pair.

Feb/Mar 2026 (IM352IA)

No.	Question	M	BT	CO
3(a)	Compare process structures in services and manufacturing.	8	4	3
3(b)	Explain long-term capacity planning. Describe capacity timing strategies and capacity sizing strategies with suitable explanations.	8	4	3
4(a)	Explain timing and sizing strategies in capacity planning with diagrams.	8	4	3
4(b)	How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer introduced variability?	8	2	2

Feb/Mar 2025 (IM352IA)

No.	Question	M	BT	CO
3(a)	Explain the four inventory strategies that should be coordinated with process choice in a manufacturing environment. Provide example for each of the strategy.	10	3	2
3(b)	Provide a note on the three elements in a customer-contract matrix in service processes.	6	2	2
4(a)	Operations managers must examine three dimensions of capacity strategy before making capacity decisions. What are they? Explain each with suitable examples.	10	3	2
4(b)	Enumerate the four steps of systematic approach to long term capacity decisions.	6	2	2

Apr/May 2024 (21IM53)

No.	Question	M	BT	CO
3(a)	List and explain operations process characteristics. Also briefly discuss the implication of four characteristics of operations processes.	8	3	2
3(b)	How does the degree of customer contact, customization and process characteristics help in structuring processes in a service organization?	8	3	2
4(a)	Explain the concept of economies of scale when planning capacity for long term implementation.	8	2	2
4(b)	The best choice for a manufacturing process depends on the volume and degree of customization required of the process. Justify the above statement with the help of a suitable matrix.	8	3	3

Mar/Apr 2023 (18IM54)

No.	Question	M	BT	CO
3(a)	Describe the four basic process strategies.	8	–	–
3(b)	List the different tools for process analysis and design and explain them.	8	–	–
4(a)	Define strategic fit. Describe the elements of strategic fit.	8	–	–
4(b)	What is the Six Sigma model of quality improvement? Describe the five steps of the Six Sigma improvement model.	8	–	–

Jan/Feb 2021 (18IM54)

No.	Question	M	BT	CO
3(a)	How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer-introduced variability?	6	–	–
3(b)	A work sampling study was conducted to determine the proportion of the time a worker is idle. The following information was gathered on a random basis.	10	–	–

Day	Number of times worker idle	Total number of observations
Monday	17	44
Tuesday	18	56
Wednesday	14	48
Thursday	16	60

(i) Based on these preliminary results, what percent of the time is the worker working? (ii) If the idle time is judged to be excessive, what additional categories might you add to a follow-up work sampling study to identify the root causes?

4(a)	Performance criteria important in creating a layout can go well beyond communication and materials handling. Identify the types of layout performance criteria that might be most important in the following settings: (i) Airport (ii) Bank (iii) Classroom (iv) Product designers' office.	8	–	–
4(b)	Vera Johnson and Merris Williams manufacture vanishing cream. Their packaging process has four steps: (1) mix, (2) fill, (3) cap, and (4) label. They have had the reported process failures analyzed, which shows the following (frequency out of 60 total failures): lumps of unmixed product – 7; over or under filled jars – 18; jar lids did not seal – 6; labels rumpled or missing – 29. *Source: the paper gives this data without a separate instruction line*	8	–	–

Jan/Feb 2021 (16IM55)

No.	Question	M	BT	CO
3(a)	Identify and discuss with block diagram the important four decisions that are essential for an effective process design.	8	–	–
3(b)	Identify and discuss the strategies used for structuring the process in manufacturing.	8	–	–
4(a)	The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes. The choices should fit the situation and each other. When the fit is more strategic, the process will be more effective. As industrial engineer you are required to identify, examine and discuss the services and manufacturing processes, looking for ways to test for strategic fit.	8	–	–
4(b)	Identify and discuss the key elements of process reengineering.	8	–	–

Fast Track Oct 2020 (16IM55/12IM62)

No.	Question	M	BT	CO
3(a)	Explain why processes exist everywhere in all organizations. Take the following supply chain activities and give a brief description for each: (i) Warehousing (ii) Customer service (iii) Logistics.	8	–	–

No.	Question	M	BT	CO
3(b)	Briefly discuss on the process strategy decisions. Use a block diagrammatic representation to bring out the connections among major process strategy decisions for effective processes.	8	–	–
4(a)	Define the following terms clearly: (i) Customer increment (ii) Resource flexibility (iii) Capital intensity (iv) Economies of scope.	8	–	–
4(b)	Position a process on the customer-contact matrix or product process matrix. Bring out the characteristics of the following types of process choices: (i) Job processes (ii) Batch processes (iii) Line processes (iv) Continuous flow processes.	8	–	–

Nov/Dec 2019 (16IM55)

No.	Question	M	BT	CO
3(a)	Identify and discuss with block diagram the important four decisions that are best understood as an effective process design.	8	–	–
3(b)	Identify and discuss the strategies used for structuring the process in manufacturing.	8	–	–
4(a)	The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes. The choices should fit the situation and each other. When the fit is more strategic, the process will be more effective. As an industrial engineer you are required to identify, examine and discuss the services and manufacturing processes, looking for ways to test for strategic fit.	8	–	–
4(b)	Identify and discuss the key elements of process reengineering.	8	–	–

Fast Track Jul 2019 (16IM55/12IM62)

No.	Question	M	BT	CO
3(a)	Construct the product-process matrix and show in this matrix the process characteristic categories, level of product customization and the elements of volume. Elaborate on the type of manufacturing process structure that exist in various types of organization briefly.	8	–	–
3(b)	"After a process is documented, metrics data collected and disconnects identified, the process analyst or design team puts together a set of changes that will make the process better". Compare this with a doctor diagnosing the patient and recommending treatments based on the diagnosis. Explain how questioning and brainstorming will assist the process designer in analyzing the gaps in the process. Take an example to support your answer.	8	–	–
4(a)	Develop the customer-contact matrix and indicate the elements of degree of customer contact, customization and process characteristic. Elaborate on the process divergence and flow based on the customer-contact matrix. How does this matrix provide the starting point for evaluating and improving a process?	8	–	–
4(b)	Capital intensity is the mix of equipment and human skills in the process. Carefully examine the advantages and disadvantages of automating a manufacturing process. Distinguish between fixed automation and flexible automation. Give two examples for fixed and flexible automation respectively. Justify automation in the context of capital intensity.	8	–	–